Transformer

Parameter	Requirement
Turns ratio	1:8 to 1:10 (step-up)
Output voltage	1.0 – 1.5 kV
Frequency range	100 Hz – 10 MHz
Secondary config	Center-tapped, 180° outputs
Load type	Capacitive, ~1 nF per output
Isolation voltage	≥ 2 kV

Commercial Transformer Research

No commercial part met the full specification. The fundamental obstacle is a three-way conflict between turns ratio, output voltage, and frequency range. RF wideband transformers achieve the frequency range but are limited to signal-level voltages (≤ 500 V) and current ratings under 250 mA. Audio transformers like Lundahl achieve the 1:20 turns ratio and high isolation voltage (3 kV) but roll off above 100 kHz. Power transformers like the Hammond 740 reach the correct voltage and turns ratio (1:20.8, 2.4 kV center-tapped) but operate only at 50/60 Hz line frequency because their silicon-steel cores saturate above a few hundred Hz. Pulse transformers approach the voltage and frequency targets but lack the required turns ratio and center-tap configuration.

Conclusion: Custom winding was the most viable path forward for the most ideal transformer that best fit our constraints.

Core Material/Geometry Selection

Toroid: Material 43 (NiZn ferrite) is electrically insulating and rated for impedance transformer use from 1–50 MHz, making it suitable for continuous RF operation.

U Shaped Brackets: Material 77 (MnZn ferrite) offers higher permeability but is electrically conductive. At MHz frequencies, eddy currents within the core itself cause severe losses.

Toroids are preferred over U-core because the closed magnetic path minimizes leakage inductance, which directly determines the upper frequency limit when driving a capacitive load. Larger toroids are preferred because a bigger cross-sectional area (Ae) reduces flux density at low frequencies, preventing core saturation and allowing operation down to 100 Hz.

Turn Count Calculations

Minimum primary turns to avoid core saturation: $N_{min} = V / (4.44 \times f \times B_{max} \times Ae)$, evaluated at the lowest operating frequency and maximum primary voltage. Secondary turns are set by the 1:5 ratio per half (1:10 total center-tapped).

Custom Winding Tests

Next Steps

Wind FT-240-43 (61mm): 65 primary turns (AWG 22), 325 turns per secondary half Theoretical lower cutoff ~86 Hz which is the best candidate for full operating range.

Continue small toroid testing: Retest with PA107 as low-impedance source, find exact SRF with open-circuit secondary, verify phase balance between center-tapped outputs.

Integrate with PA107: Run full test suite on wound FT-240-43, such as frequency response, phase balance, 1 nF capacitive load test, and incremental high voltage ramp to 1.5 kV.